Weekend patch waves with Azure Arc and Update Manager: one overview, four waves, no redundant pair down at once

A practical design for Arc-enabling servers on any platform (on-premises Windows and Linux, VMware, AWS, GCP and other clouds) and patching them with Azure Update Manager in a Friday-to-Monday wave model based on criticality, sensitivity and production dependencies: a tag model enforced with Azure Policy, governed onboarding, one Resource Graph overview of every server, maintenance configurations per wave with dynamic scopes, pre-event health gates so Monday never starts if Sunday's side A failed, a check that flags redundant pairs sharing a wave, emergency patching, reporting, rollback and cost. Verified against Microsoft Learn; practical, not lab-tested.

7 phases | 21 steps | Tick each item when its "done when" criterion is true

Phase 1 Design the wave and tag model about 2–5 days (mostly owner workshops)

- [ ] **1.1 Fix a small tag vocabulary**
  A one-page tag standard lists names and allowed values: `patch-wave` (fri | sat | sun | mon | oob), `criticality` (low | medium | high | critical), `sensitivity` (e.g. public | internal | confidential | restricted), `redundancy-group` (free text, one value per service pair), `redundancy-side` (A | B), `owner` (team mailbox or group)
- [ ] **1.2 Classify servers with owners, not for them**
  Each application owner has confirmed wave, criticality, sensitivity and any redundancy pair for their servers, and the result is in a version-controlled file or CMDB
- [ ] **1.3 Anchor the four waves to Patch Tuesday**
  Each wave has a recurrence, start time, duration and time zone written down, and a pilot or test step precedes Friday

Phase 2 Enforce the tags with Azure Policy about 1–2 days

- [ ] **2.1 Require the tags with built-in policies**
  Built-in 'Require a tag on resources' assignments exist for `patch-wave`, `criticality` and `owner` on the Arc landing-zone scope; compliance is reviewed weekly
- [ ] **2.2 Add a custom rule for allowed waves and the A/B rule**
  A custom policy (Audit first, Deny later) flags any Arc server or Azure VM whose `patch-wave` is outside the allowed list, or whose `redundancy-side` contradicts its wave
- [ ] **2.3 Enforce periodic assessment on the same scope**
  The built-in periodic-assessment policies for Azure Arc-enabled servers (and Azure VMs) are assigned with remediation, separately for Windows and Linux

Phase 3 Arc-enable servers on every platform into a governed landing zone about 1–10 days depending on estate

- [ ] **3.1 Prepare one governed landing zone for Arc machines**
  A dedicated subscription or resource groups per site/platform exist with resource providers registered, the policies from phase 2 assigned, and an extension allowlist decided
- [ ] **3.2 Use a least-privilege onboarding identity**
  A service principal with only the Azure Connected Machine Onboarding role, scoped to the target resource groups, is used for onboarding; its secret lives in a vault and has an expiry
- [ ] **3.3 Pick the onboarding path per platform**
  Each platform has a named method and owner: on-premises physical/Hyper-V, VMware vSphere, AWS, GCP and other clouds

Phase 4 Build the one overview: Resource Graph and a workbook about 1–2 days

- [ ] **4.1 Create the estate overview query**
  The query returns one row per machine across all subscriptions in scope; machines with no assessment or no run are visible, not filtered out
- [ ] **4.2 Add the redundant-pair check**
  A second query lists every redundancy group whose members share a wave, have only one side tagged, or have a side in the wrong wave; the list is empty before each weekend

- [ ] **4.3 Publish both in a workbook**
  A shared workbook shows the overview, the pair check and per-wave totals; application owners have Reader access to it

Phase 5 Maintenance configurations per wave with dynamic scopes about 1–2 days

- [ ] **5.1 Create one configuration per wave**
  `mc-wave-fri`, `mc-wave-sat`, `mc-wave-sun` (or `sun-1` / `sun-2`) and `mc-wave-mon` exist with scope `InGuestPatch`, the agreed recurrence, reboot setting and classifications, deployed from source control
- [ ] **5.2 Attach machines with tag-driven dynamic scopes**
  Each configuration has a dynamic scope over all Arc and workload subscriptions filtering only on its `patch-wave` value; the scope preview matches the overview query for that wave

Phase 6 Health gates: Monday only starts when Sunday side A is healthy about 2–3 days

- [ ] **6.1 Wire pre and post events**
  `mc-wave-sun` and `mc-wave-mon` have Event Grid subscriptions for `Microsoft.Maintenance.PreMaintenanceEvent` and `Microsoft.Maintenance.PostMaintenanceEvent`, delivered to a runbook webhook or Azure Function with a managed identity
- [ ] **6.2 Implement the Monday gate**
  Monday's pre-event runs the gate query plus Sunday's recorded service checks, and calls the cancellation API when any side-A machine failed, is disconnected or has no result
- [ ] **6.3 Add lighter gates for Saturday and Sunday**
  Saturday and Sunday pre-events check that the previous wave has no failed runs above an agreed threshold before starting

Phase 7 Emergency patching, reporting, rollback and cost about 1–2 days, then monthly

- [ ] **7.1 Patch out of band without breaking the pair rule**
  An emergency runbook exists: one-time updates on side A and non-redundant servers first, verification, then side B — never both sides in one action
- [ ] **7.2 Report per wave and alert on exceptions**
  After each weekend a report shows per wave: machines in scope, succeeded, failed, not run, pending critical/security; alerts go to the `owner` team
- [ ] **7.3 Roll back: stop the next wave first**
  For a bad update the team can, in order: cancel the next wave, exclude the KB/package in later waves, uninstall or restore on affected machines, verify
- [ ] **7.4 Budget Arc servers in Update Manager from Microsoft's rules**
  A cost line lists Arc servers that are charged and those that are not, using Microsoft's FAQ and pricing page

Owner	Change / ticket	Date completed

Original checklist by Tenant Recipes, an independent project not affiliated with Microsoft or any vendor. Every step, its actions, pitfalls and the official sources it was checked against are on https://tenantrecipes.com/guides/arc-update-manager-weekend-waves.html. Time estimates are our own planning figures. Confirm against the linked sources and test in a pilot before production.